

Production Order List — Detailed Design Document (詳細設計書)

002_DD — implements 002_BD (SCR-002), requirements REQ-020–REQ-027.

Document control (改版履歴)

Field	Value
Document ID	002_DD
Category	UI
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-003
Implements	002_BD
Created by	Claude (for ThanhTN)
Created date	2026-09-22
Last updated by	Claude (for ThanhTN)
Last updated date	2026-09-23

Version	Date	Author	Revision content
1	2026-09-22	Claude (for ThanhTN)	Initial creation
2	2026-09-22	Claude (for ThanhTN)	Rendered mockup published and linked
3	2026-09-22	Claude (for ThanhTN)	Aligned with the implementation: string-bound query parameters, enum names matched by name only, the sort applied before the projection, and the two contrast/link-style corrections axe found
4	2026-09-23	Claude (for ThanhTN)	Diagrams redrawn for the revised DD template (RFC 0009): the layout sketch and the empty/no-match states as SVG wireframes under <code>wireframes/</code> ; the screen states also drawn as a Mermaid state diagram above the table. No field, rule, API or processing changes

Version	Date	Author	Revision content
5	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. The message catalog lists the Japanese text with an English gloss (WI-005 DEC-003); table caption, summary, button names, formatting helpers, wireframes and mockup updated. Test-viewpoint rows keep their English shorthand for button names. A Japanese edition of the mockup was added (mockups/002_DD_screen-b-mockup.ja.html). No field, rule, API or processing changes

Overview and reference documents (概要・目次)

Field	Value
File / component name	<code>ProductionOrderListPage.tsx</code> and its child components; <code>ProductionOrdersController.List</code>
Overview	SCR-002, the production-order list: a read-only, server-paged, sortable, filterable table whose whole view state lives in the URL, and from which an order is opened in SCR-001

Module / method / processing index

No	Name	Overview	Notes
1	<code>ProductionOrderListPage</code>	Route component: view state, query, state selection	Module design §1
2	<code>ProductionOrderFilters</code>	Filter panel, client validation, Search/Clear	§2
3	<code>ProductionOrderTable</code>	Results table / SP cards, sorting, row activation	§3
4	<code>ListPagination</code>	Summary, page size, Previous/Next	§4
5	<code>ProductionOrderListQuery</code> + <code>validateFilters</code>	The query validator — one rule set, two implementations (server authoritative)	§5
6	<code>useListViewState</code>	URL ⇄ typed view state	§6
7	<code>ProductionOrdersController.List</code>	Request boundary	§7
8	<code>api.ts</code> — <code>listOrders</code>	Typed client wrapper	§8
9	<code>ProductionOrderService.ListAsync</code> and repository read methods	Server-side query use case	Pointer only — designed in 002_DD-FN

Reference documents

No	Document	Purpose / use	Notes
1	002_BD	Screen items, validation V-09–V-13, events E-10–E-19, states	Parent
2	002_DB	Indexes, query shape, sort mapping, demo seed	Data layer
3	002_DD-API	Endpoint contract	Companion
4	002_DD-FN	Service and repository methods	Companion
5	002_DD-SPD	Step-by-step processing per component	Companion
6	001_DD, 001_DD-API, 001_DD-FN, 001_DD-SPD	SCR-001: message catalog, error contract, <code>Result<T></code> , <code>ApiClient</code> , plant clock, telemetry — all reused	Screen A
7	0001_ADR, 0002_ADR	Layered backend; cookie auth and roles	
8	001_BD	SCR-001's screen transition, which this screen changes (Cancel target)	

Referenced by

No	Document	Purpose / use	Notes
1	002_DD-API, 002_DD-FN, 002_DD-SPD	Companions elaborating this DD	
2	<code>work-items/WI-003/test-plan.md</code>	Test viewpoints below become <code>TC-###</code>	Not yet written

Component / file organization

X-1. Path structure

No	Path / namespace	Purpose	Notes
1	<code>src/frontend/src/features/production-orders/</code>	Existing feature folder; the list page and its components join SCR-001's files	<code>ProductionOrderListPage.tsx</code> , <code>ProductionOrderFilters.tsx</code> , <code>ProductionOrderTable.tsx</code> , <code>ListPagination.tsx</code> , <code>useListViewState.ts</code> , <code>listValidation.ts</code>
2	<code>src/frontend/src/features/production-orders/api.ts</code> , <code>types.ts</code> , <code>messages.ts</code>	Existing files, extended — not duplicated	<code>listOrders</code> , the list types, the new message IDs
3	<code>src/backend/ProductionManagementAI.Api/Controllers/ProductionOrdersController.cs</code>	Existing controller, one new action	

No	Path / namespace	Purpose	Notes
4	src/backend/ProductionManagementAI.Application/ProductionOrders/	ProductionOrderListQuery , ProductionOrderListContracts , ListAsync on the existing service	
5	src/backend/ProductionManagementAI.Infrastructure/ProductionOrders/	Repository read methods, query extensions	
6	src/backend/ProductionManagementAI.Infrastructure/Migrations/	The two migrations from 002_DB	

X-2. Shared/common components used

No	Name	Purpose	Notes
1	apiClient	Fetch wrapper with the 401 → /login behavior	WI-001
2	AppHeader , ProtectedRoute	Common header and route guard	WI-001
3	MessageBanner	Error/success banner with its live region	001_DD module 8
4	messages.ts , statusLabels	The single message catalog and status labels	001_DD

X-3. Feature-level components used

No	Name	Purpose	Notes
1	ProductionOrderFilters	Filter panel	\$2
2	ProductionOrderTable	Results	\$3
3	ListPagination	Paging controls	\$4

X-4. External APIs used — see "APIs used" below.

X-5. Responsive composition — one set of components with responsive Tailwind styling, not separate PC/SP components, except the results region:

`ProductionOrderTable` renders a `<table>` at `sm` and above and a card list below it, because a table and a card list cannot be the same markup and stay semantically correct. The filter panel is the same markup, wrapped in a `<details>`-style disclosure below `sm`.

Companion design documents

No	Document	Type	Covers
1	002_DD-API_製造指示一覽.md	api-design	<code>GET /api/production-orders</code> (new, API-PO-04) and the reuse of <code>GET /api/products</code> ; request/response field catalogs, error codes
2	002_DD-FN_製造指示一覽.md	function-design	<code>ProductionOrderService.ListAsync</code> , the repository's <code>CountOrdersAsync</code> / <code>ListOrdersAsync</code> , filter/sort composition, the list mapper, observability
3	002_DD-SPD_製造指示一覽.md	screen-processing-design	P-10 to P-15 plus the controller request boundary, one block per component

Task / design index

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
1	basic-design	002_BD	SCR-002	—	yes	—	ThanhTN	no	2026-09-22	Approved ("the BD look good"); revised to v2/v3 during 002_DB and 002_DD
2	database-design	002_DB	Indexes, seed, queries	—	yes	—	ThanhTN	no	2026-09-22	Approved ("the DB design is approved")
3	detailed-design	002_DD	Modules §1–§8, states, mockup	—	no	—	ThanhTN	no	2026-09-22	This document; mockup published 2026-09-22
4	api-design	002_DD-API	<code>GET /api/production-orders</code>	Query parameters, error codes	no	—	ThanhTN	no	2026-09-22	
5	function-design	002_DD-FN	<code>ListAsync</code> , repository reads	Sort mapping, overdue rule	no	—	ThanhTN	no	2026-09-22	
6	screen-processing-design	002_DD-SPD	P-10–P-15	Response handling	no	—	ThanhTN	no	2026-09-22	

Module design

1. ProductionOrderListPage

Field	Value
Description	Route component for <code>/production-orders</code> : owns the query lifecycle and chooses which of the six screen states to render
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: rendered inside `ProtectedRoute` , so a session exists.

Not applicable — plain component, no rule/validator fields.

Arguments: none (route component).

Dependencies

No	Module / component	Overview	Notes
1	<code>useListViewState</code>	View state from the URL	§6
2	<code>api.ts</code> <code>listOrders</code> , <code>listProducts</code>	Data	§8
3	<code>ProductionOrderFilters</code> , <code>ProductionOrderTable</code> , <code>ListPagination</code> , <code>MessageBanner</code>	Children	§2–§4
4	<code>useAuth</code>	Role gate (UX only; the server is authoritative)	WI-001

Processing overview: see 002_DD-SPD §1 and §6. One query per distinct view state; a superseded response is discarded rather than rendered.

Processing flow: 002_DD-SPD §1 (P-10) and §6 (P-15).

Return value

Type	Name	Description
JSX	—	One of <code>loading</code> , <code>ready</code> , <code>empty</code> , <code>noMatch</code> , <code>error</code> , <code>forbidden</code>

2. ProductionOrderFilters

Field	Value
Description	The filter panel (items 7–13): holds draft filter values and publishes them on Search
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: the product options have loaded, or the product select is rendered disabled with its load error.

Not applicable — plain component; its rules live in §5.

Arguments

No	Type	Name	Description
1	<code>ListViewState</code>	<code>view</code>	Current applied filters, used to (re-)initialize the draft
2	<code>ProductResponse[]</code>	<code>products</code>	Options for item 8
3	<code>(filters) => void</code>	<code>onApply</code>	Publishes valid filters to the view state
4	<code>boolean</code>	<code>busy</code>	Disables Search while a query is in flight

Dependencies

No	Module / component	Overview	Notes
1	<code>validateFilters</code>	Client-side V-09, V-10	§5
2	<code>messages.ts</code>	ID → text	

Processing overview: 002_DD-SPD §3 (P-12). The draft re-bases whenever the applied view state changes, so Back or a shared URL always leaves the panel showing what produced the rows.

Return value: JSX (a `<form>`).

3. ProductionOrderTable

Field	Value
Description	The results region (items 16–22, 27): a table at <code>sm</code> and above, a card list below it
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: called only in the `ready` state, with `items.length ≥ 0`.

Not applicable — plain component.

Arguments

No	Type	Name	Description
1	<code>ProductionOrderListItem[]</code>	<code>items</code>	The page's rows
2	<code>ProductionOrderSort</code>	<code>sort</code>	Active sort key, from the response
3	<code>'asc' \ 'desc'</code>	<code>dir</code>	Active direction, from the response
4	<code>(sort, dir) => void</code>	<code>onSort</code>	Publishes a new sort to the view state

Dependencies

No	Module / component	Overview	Notes
1	<code>Link</code>	Order-number cell → SCR-001	react-router
2	<code>statusLabels</code> , <code>labels</code> , date and number formatting	M-05, M-10	<code>messages.ts</code> ; <code>src/lib/format.ts</code> (WI-005)

Processing overview: 002_DD-SPD §4 (P-13). Accessibility specifics: `<caption>` naming the table and its sort, 「製造指示一覧（納期の昇順）」 (Production orders, sorted by due date ascending), `<th scope="col">` header cells each containing a `<button>`, `aria-sort="ascending" | "descending"` on the active header only, the overdue marker as text next to the due date, and the row's mouse handler deliberately additive to the link.

Return value: JSX.

4. ListPagination

Field	Value
Description	Result summary (item 14), page-size select (15) and Previous/Next (23)
Return type	JSX element
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: a response with `total`, `page` and `pageSize` exists.

Not applicable — plain component.

Arguments

No	Type	Name	Description
1	<code>number</code>	<code>total</code> , <code>page</code> , <code>pageSize</code>	From the response, not the request
2	<code>(page) => void</code> / <code>(size) => void</code>	<code>onPage</code> , <code>onPageSize</code>	Publish to the view state

Dependencies: none beyond \$6.

Processing overview: 002_DD-SPD \$5 (P-14). The summary is the screen's live region: when a query finishes, it announces 「{total}件中 {first}～{last}件」 ({first}–{last} of {total} orders) (M-09).

Return value: JSX.

5. ProductionOrderListQuery (server) and validateFilters (client)

Field	Value
Description	The list query's validator and normalizer: one rule set stated once here, implemented on both sides, with the server authoritative
Return type	<code>Result<ProductionOrderListQuery></code> (server) / <code>Record<field, messageId></code> (client)
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: none — it is a pure function of the raw query values.

Rule type: input validation plus normalization (format, range, cross-field and allow-list checks).

Rule references

No	Rule / validator	Location	Notes
1	002_BD V-09–V-13	002_BD §5	The rules this implements
2	ProductExistsAsync	002_DD-FN §1 step 2	The one check that needs the database, so it runs in the service, not here

Condition / data references

No	Key	Source	Notes
1	Allowed statuses	ProductionOrderStatus enum	Domain
2	Allowed sort keys / directions	ProductionOrderSort , SortDirection enums	Defined with this module
3	Allowed page sizes	{10, 20, 50, 100} constant	DEC-002

Check parameters

No	Name	Type	Target field
1	status	string[]	Status filter (V-11)
2	productId	string	Product filter (V-12, existence in the service)
3	dueFrom , dueTo	string (date)	Due-date range (V-10)
4	orderNumber	string	Fragment (V-09)
5	sort , dir , page , pageSize	string / int	Sort and paging (V-13)

Arguments

No	Type	Name	Description
1	raw query values	request	Query-string values (server) or draft filter values (client)

Dependencies: none — deliberately pure, so both sides can unit-test it without a database or a DOM.

Processing overview: validate, then normalize. Absent values take defaults; present-but-invalid values are rejected with their message ID and never silently defaulted (002_BD V-13) — that asymmetry is the whole point of the module, because defaulting a bad value would let a crafted request widen the result set. Normalization

covers the order-number fragment (trim → length check → upper-case → escape `\`, `%`, `_` → wrap in `%...%`) and de-duplicating the status set. Every offending parameter is reported in one response, not the first one only.

The client implements V-09 and V-10 only (the two the user can type wrong); V-11, V-12 and V-13 cannot be violated through the UI, since those values come from the screen's own controls, so the client does not re-implement them — the server still enforces all five.

Processing flow: 002_DD-SPD §7 step 3 (server) and §3 step 3 (client).

Return value

Type	Name	Description
<code>Result<ProductionOrderListQuery></code>	server	<code>Ok(query)</code> or <code>Invalid(parameter → message ID)</code>
<code>Record<string, string></code>	client	Empty when valid; field → message ID otherwise

6. useListViewState

Field	Value
Description	Hook converting between the URL query string and the typed view state
Return type	<code>[ListViewState, (next, options) => void]</code>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: rendered inside the router.

Not applicable — plain hook.

Arguments: none.

Dependencies

No	Module / component	Overview	Notes
1	<code>useSearchParams</code>	URL access	react-router

Processing overview: 002_DD-SPD §2 (P-11). Parsing is total and never throws; serialization omits every parameter equal to its default.

Return value

Type	Name	Description
[ListViewState, setView]	—	Current view state and the publisher that rewrites the URL

7. ProductionOrdersController.List

Field	Value
Description	GET /api/production-orders — the request boundary
Return type	ActionResult<PagedResult<ProductionOrderListItem>>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: the existing ProductionOrderEditor policy has authorized the request.

Not applicable — plain handler.

Arguments

No	Type	Name	Description
1	ProductionOrderListRequest	request	[FromQuery] binding of the nine parameters
2	CancellationToken	cancellationToken	

Dependencies

No	Module / component	Overview	Notes
1	ProductionOrderListQuery.TryCreate	Validation	\$5
2	ProductionOrderService.ListAsync	Use case	002_DD-FN \$1
3	ProductionOrderProblems.ToActionResult	Result<T> → HTTP	Existing

Processing overview: 002_DD-SPD \$7. No [Consumes] attribute — the endpoint takes no body.

Return value: 200 with the paged result, or the Problem Details responses catalogued in 002_DD-API.

8. `api.ts` — `listOrders`

Field	Value
Description	Typed wrapper building the query string and parsing the paged response
Return type	<code>Promise<PagedResult<ProductionOrderListItem>></code>
Created by / date	Claude / 2026-09-22
Last modified by / date	—

Preconditions: none.

Not applicable — plain function.

Arguments

No	Type	Name	Description
1	<code>ListViewState</code>	<code>view</code>	Serialized with the same rules as the URL (§6), so what the user sees and what is queried cannot diverge
2	<code>AbortSignal</code>	<code>signal</code>	Cancels a superseded query

Dependencies: `ApiClient` (401 handling, JSON parsing, Problem Details mapping).

Processing overview: one serializer for both the URL and the request; defaults are omitted from both.

Return value: the parsed `PagedResult`, or a typed API error the page maps per 002_DD-SPD §6.

9. Server-side service and repository methods

Pointer only — `ProductionOrderService.ListAsync`, `IProductionOrderRepository.CountOrdersAsync` / `ListOrdersAsync`, the filter/sort composition and the list mapper are designed in **002_DD-FN**, which also specifies the spans and metrics for the endpoint.

Screen layout and mockup

Implementation-fidelity layout. It refines 002_BD §1 in two ways: the filter panel is laid out as a labelled grid that reflows from four columns to one, and the overdue marker is specified as a text badge after the due date rather than a styling of the date itself, so it survives both themes and screen readers.

1 ProductionManagementAI

ダッシュボード

製造指示一覧

+ 新規製造指示

2 Admin · ログアウト

3 ホーム > 製造指示一覧

4 製造指示一覧

5 + 新規製造指示

6 Only in the error state: 「問題が発生しました。もう一度お試しください。」

再試行

絞り込み

7 ステータス

☒ 下書き

☒ 進行中

☐ 完了

☐ 取消

8 製品

すべての製品

9 納期（開始）

2026/09/01

error text slot

10 納期（終了）

2026/10/31

error text slot

11 指示番号

00042

error text slot

12 クリア

13 検索

14 87件中 1～20件

15 表示件数

20

16 指示番号 ▲	17 製品	18 数量	19 納期	20 ステータス	21 更新日時
22 PO-2026-00042	P-1004 — ドライブシャフト	250	2026/09/18	納期遅れ 27 進行中	2026/09/20 10:12
PO-2026-00043	P-1001 — ブレーキキャリパー	120	2026/09/30	下書き	2026/09/19 14:03
...					

23 < 前へ

1 / 5 ページ

次へ >

Filters reflow from a four-column grid to one column below 640px. The overdue marker is a text badge after the due date.

Empty state (no orders at all) and no-match state (filters set, nothing matched):

24 Empty state — no orders at all

製造指示はまだありません。最初の1件を登録してください。

+ 新規製造指示

Replaces items 14–23.

25 No-match state — filters set, nothing matched

絞り込みに一致する製造指示はありません。
納期の範囲を広げるか、製品の絞り込みを外してください。

絞り込みをクリア

The filter panel above keeps its values.

Mockup artifact: <https://claude.ai/artifact/2XrZ9xnEfzQ6pCbUnZovL3> — published privately on 2026-09-22 with the user's authorization (plan revision 1, "Resources and external actions"). Source kept in the repository at `docs/en/020_detailed-design/002/mockups/002_DD_screen-b-mockup.html` , alongside 001_DD's, whose token system and app-look CSS it reuses so the two mockups read as one set. Seven artboards: 1 default view, 2 filtered and sorted (with the focus rings), 3 filter validation errors, 4 no-match, 5 empty, 6 query error, 7 SP card layout.

Japanese edition, with the page captions in Japanese too, linked from the Japanese PDF (WI-005): <https://claude.ai/artifact/TTxzX1PJVyKr3aLYpHSRZK> (private). Source: [mockups/002_DD_screen-b-mockup.ja.html](#) .

Region	Contains (field/control)	Notes
Header	AppHeader (items 1, 2)	Shared component, unchanged
Title row	Heading (4), New production order (5)	Button right-aligned on PC, full width on SP
Banner	MessageBanner (6)	Rendered only in the <code>error</code> state; <code>role="alert"</code> , carries 再試行 (Retry)
Filter panel	Items 7–13	4-column grid at <code>lg</code> , 2 at <code>sm</code> , 1 below; each field has a reserved error slot so the layout doesn't jump when an error appears
Summary row	Items 14, 15	Summary is the live region (<code>role="status"</code>)
Results	Items 16–22, 27	<code><table></code> at <code>sm</code> +, card list below; sticky header row on PC
Paging	Item 23	Centered; controls disabled, not hidden, at the ends
Empty / no-match	Items 24, 25	Replace the results region only; the filter panel stays

Screen item definition

Types and client-side behavior for each input; the authoritative rules are 002_BD §5 and the server (§5 above).

Field	Type	Required	Validation rule	Source (BD ref)
status	Set<'Draft'\ 'InProgress'\ 'Completed'\ 'Cancelled'>	no	Four checkboxes in a <fieldset> with a <legend>; empty set = no restriction	§3 item 7, V-11, DEC-005
productId	string \ null (uuid)	no	Native <select>, options ordered by SKU, label "{sku} — {name}" (e.g. P-1004 – ドライブシャフト), first option 「すべての製品」 (All products) with an empty value	§3 item 8, V-12, M-06
dueFrom	string \ null (YYYY-MM-DD)	no	<input type="date">; malformed → MSG-E016; after dueTo → MSG-E017	§3 item 9, V-10
dueTo	string \ null (YYYY-MM-DD)	no	<input type="date">; malformed → MSG-E016	§3 item 10, V-10
orderNumber	string \ null	no	<input type="search" maxlength="20">; trimmed on submit; longer than 20 → MSG-E015 (the attribute prevents it in practice, the check covers paste and crafted URLs)	§3 item 11, V-09
sort	ProductionOrderSort	no	Set by the header buttons only; never typed	§3 items 16–21, V-13
dir	'asc' \ 'desc'	no	Toggled by re-clicking the active header	§3 items 16–21, V-13
page	number	no	Set by the paging controls; reset to 1 by any filter, sort or page-size change	§3 item 23, V-13
pageSize	10 \ 20 \ 50 \ 100	no	<select>, default 20	§3 item 15, V-13, DEC-002

Read-only row fields (items 16–22, 27) are rendered from ProductionOrderListItem (002_DD-API) and have no input behavior. isOverdue comes from the server, so every client agrees on "today" (WI-002 DEC-011).

Message catalog additions (messages.ts; the server returns the same IDs as code; other UI strings are in its labels object, WI-005 DEC-007). 001_DD already defines MSG-E001–MSG-E014 and MSG-I001–MSG-I002, so Screen B continues the same catalog rather than starting a second one:

ID	Text (Japanese, as implemented)	English gloss
MSG-E015	指示番号の検索は20文字以内で入力してください。	Order number search can't exceed 20 characters.

ID	Text (Japanese, as implemented)	English gloss
MSG-E016	有効な日付を入力してください。	Enter a valid date.
MSG-E017	納期（開始）は納期（終了）以前の日付にしてください。	'Due from' must be on or before 'Due to'.
MSG-E018	不明なステータスです。	Unknown order status.
MSG-E019	対応していない並べ替えまたはページングの指定です。	Unsupported sort or paging option.
MSG-E020	製造指示を閲覧する権限がありません。	You don't have permission to view production orders.
MSG-I003	製造指示はまだありません。最初の1件を登録してください。	No production orders yet. Create the first one.
MSG-I004	絞り込みに一致する製造指示はありません。	No orders match your filters.

Reused unchanged: MSG-E002 (unknown product filter), MSG-E013 (query failure).

Two visual rules were corrected during implementation, after `@axe-core/playwright` failed on the real screen: the `Cancelled` badge uses `text-gray-600` rather than `text-gray-400` (2.9:1 on white fails WCAG 1.4.3), and the breadcrumb link is underlined always, not only on hover (WCAG 1.4.1, "link in text block").

Loading / empty / error / success states

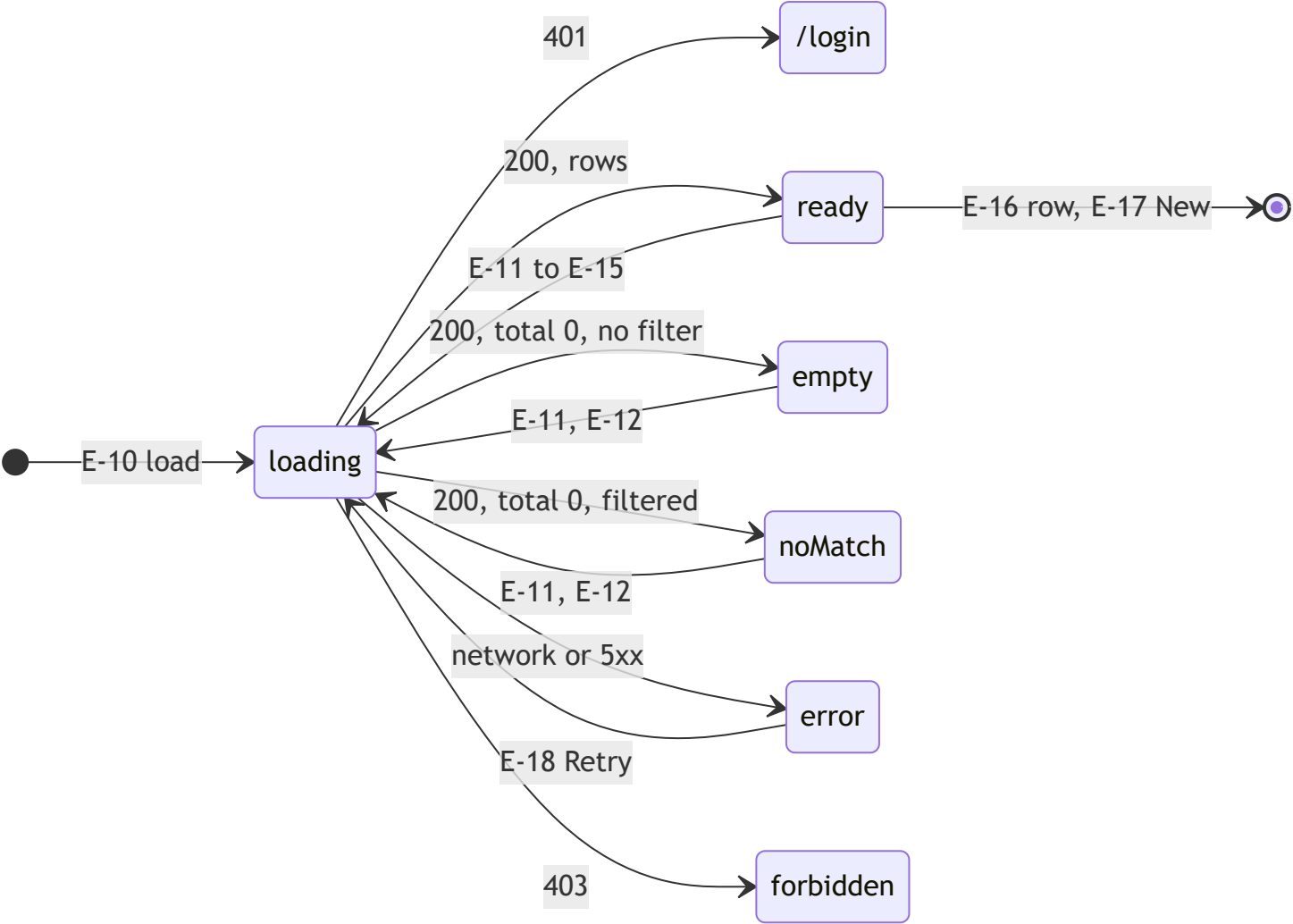
State	Trigger	UI behavior	Data shown
<code>loading</code>	A query is in flight	The results region is replaced by a skeleton with <code>aria-busy="true"</code> ; the filter panel stays usable but 検索 (Search) is disabled; previous rows are not left visible as if they were the answer	Skeleton rows
<code>ready</code>	200 with <code>total > 0</code>	Table, summary, sort marker and paging, all rendered from the response's echoed controls; the summary's live region announces the new total	The page's rows
<code>ready</code> (past the last page)	200 with <code>total > 0</code> and <code>items = []</code>	Empty table body with the paging controls intact and < 前へ (Previous) enabled — not an error (REQ-024)	No rows
<code>empty</code>	200 with <code>total = 0</code> and no filter set	MSG-I003 with a + 新規製造指示 (New production order) action, replacing the results region	—
<code>noMatch</code>	200 with <code>total = 0</code> and at least one filter set	MSG-I004 with 絞り込みをクリア (Clear filters); the filter panel keeps its values so the user can adjust rather than retype	—
Filter validation error	Client V-09/V-10, or a 400 from the server	Inline error under each offending field, focus moved to the first one; the previously loaded rows stay on screen and are not implied to match the rejected filters	Previous rows

State	Trigger	UI behavior	Data shown
error	Network failure or 5xx	MSG-E013 banner with 再試行 (Retry), which re-runs the same query unchanged	Previous rows cleared
forbidden	Client role gate or 403	Full-region panel MSG-E020 with a link back to / ; no query is issued	—
401	Any request without a valid session	apiClient redirects to /login	—

Processing and state transitions

This screen has no business state machine — it changes no order data (DEC-003). Its transitions are view-state transitions.

State transitions



Transitions from any state (E-17 New, E-19 Back/Forward) and a 400 that keeps the previous state are in the table below, not the diagram.

From state	Event	To state	Side effect
(route entered)	E-10 load	loading	View state read from the URL; product options fetched
loading	200, rows	ready	Summary announced
loading	200, total = 0 , no filter	empty	—
loading	200, total = 0 , filtered	noMatch	—
loading	400	previous state + field errors	No rows replaced
loading	401 / 403	/login / forbidden	—
loading	network or 5xx	error	—
ready / empty / noMatch	E-11 Search, E-12 Clear	loading	URL updated (push), page → 1
ready	E-13 sort	loading	URL updated (push), page → 1
ready	E-14 page, E-15 page size	loading	URL updated (push); page size also resets page → 1
ready	E-16 row activated	(leaves screen)	Navigates to SCR-001 edit mode
any	E-17 New	(leaves screen)	Navigates to SCR-001 create mode
error	E-18 Retry	loading	Same query re-run
any	E-19 Back/Forward	loading	View state re-read from the URL

Processing flows

Flow	Where (screen-processing-design / function-design section)	Events
P-10 Load and query	002_DD-SPD §1	E-10, E-19
P-11 View-state synchronization	002_DD-SPD §2	E-10–E-15, E-19
P-12 Filter, search, clear	002_DD-SPD §3	E-11, E-12
P-13 Sort and row activation	002_DD-SPD §4	E-13, E-16
P-14 Page and page size	002_DD-SPD §5	E-14, E-15
P-15 Query-response handling	002_DD-SPD §6	E-10, E-18
Server query use case	002_DD-FN §1–§6	—

APIs used

Endpoint	Method	Purpose	Design doc
/api/production-orders	GET	One page of orders with the total	002_DD-API §1
/api/products	GET	Product filter options	002_DD-API §2 (defined in 001_DD-API §1)

Database and transaction mapping

Operation	Table(s)	Transaction boundary	Concurrency handling
Count matches	production_orders	None (single statement, READ COMMITTED)	Not applicable — read-only
Fetch page	production_orders joined to products	None (single statement)	Not applicable — read-only
Product filter existence check	products	None	Not applicable

Deliberate exception to the one-query-per-table convention: the page query joins `products`, because the list must show the product label and can sort by it, and issuing a second query per page to resolve 20 product ids would be slower and could tear (a product renamed between the two queries). The count query does **not** join, since no filter needs it. Both select only the columns the screen uses (002_DB read projection). No existing SCR-001 query is modified.

The two statements can observe different snapshots, so the total may differ from the page by a row created between them; 002_DB accepts that explicitly rather than taking a repeatable-read transaction for an informational screen.

Exception handling

Failure	Retry policy	Timeout	User-facing error	Logging
Invalid query parameter (400)	none automatic — the user corrects it	—	Inline field error, or a banner for a parameter with no field	Information: rejected parameter names with their message IDs (structured)
Unknown product filter (400)	none	—	Inline error MSG-E002 on the product field	Information
Not signed in (401)	none	—	Redirect to <code>/login</code>	Existing <code>apiClient</code> behavior
Missing role (403)	none	—	Panel MSG-E020	Information

Failure	Retry policy	Timeout	User-facing error	Logging
Query/database failure (500)	none automatic; 再試行 (Retry) re-runs it	Npgsql command timeout 30s (default)	Banner MSG-E013	Error with exception, server-side only; the response is generic Problem Details with no stack trace or exception text (ai/rules/backend.md)
Frontend network failure	none automatic; 再試行 (Retry)	browser default	Banner MSG-E013	<code>console.error</code> in development only
Superseded query (user changed the view mid-flight)	not an error	—	none	none — the response is discarded by <code>AbortSignal</code>
Product options fail to load	none automatic	—	Product select disabled with an inline load error; the rest of the screen still works	Information

Test viewpoints and unresolved decisions

U = unit, I = integration, E = E2E. These become `TC-###` when the WI-003 test plan is written (plan revision 2).

Scenario	Precondition	Expected result	Test-plan ID
Default view (REQ-020) — I, E	Seeded 80 orders	200; 20 rows; <code>total = 80</code> ; sorted by due date ascending; every status present (DEC-006)	TC-101
Row content (REQ-021) — U, E	One known order	Row shows order number, SKU + name, quantity, due date, status label, updated date; no GUID and no raw enum name visible	TC-102
Overdue marker (REQ-021) — U	Plant today = D	<code>dueDate < D</code> with <code>Draft / InProgress</code> → <code>isOverdue = true</code> ; same date with <code>Completed / Cancelled</code> → <code>false</code> ; <code>dueDate = D</code> → <code>false</code>	TC-103
Status filter, multi-select (REQ-022) — I	Seeded data	<code>?status=Draft&status=InProgress</code> returns exactly those rows and a matching total; no <code>status</code> returns everything	TC-104
Product / due-range / fragment filters (REQ-022) — I	Seeded data	Each filter alone returns exactly the matching rows; range inclusive at both ends; fragment matches case-insensitively mid-string	TC-105
Combined filters (REQ-022) — I	Seeded data	Filters combine with AND; the total matches the filtered row count	TC-106
Invalid filter values (REQ-022) — I	—	Unknown status → 400 MSG-E018; unknown product → 400 MSG-E002; <code>dueFrom > dueTo</code> → 400 MSG-E017; malformed date → 400 MSG-E016; 21-character fragment → 400 MSG-E015; all offending parameters reported together	TC-107

Scenario	Precondition	Expected result	Test-plan ID
Fragment escaping (REQ-022, DEC-010) — I	An order <code>P0-2026-00042</code> exists	Searching <code>%</code> or <code>_</code> returns no rows (treated literally), not every row	TC-108
Sorting (REQ-023) — I, E	Seeded data	Each of the six keys sorts the whole result set, not just the page; clicking again reverses; status sorts in workflow order; unsupported key → 400 MSG-E019	TC-109
Paging stability (REQ-023, REQ-024) — I	Orders sharing a due date	Pages 1 and 2 together contain each row exactly once (the order-number tie-breaker)	TC-110
Paging and page size (REQ-024) — I, E	80 orders	<code>pageSize</code> 10/20/50/100 accepted, default 20; page beyond the last → 200 with <code>items = []</code> and the real <code>total</code> ; <code>pageSize=101</code> , <code>0</code> , <code>abc</code> , <code>page=0</code> → 400 MSG-E019	TC-111
Page reset (REQ-024) — U, E	On page 3	Changing a filter, the sort or the page size returns to page 1	TC-112
Navigation (REQ-025) — E	Rows present	Clicking a row, and activating the order-number link by keyboard, both open <code>/production-orders/{id}</code> ; New production order opens <code>/production-orders/new</code> ; the list exposes no save or status control	TC-113
Authorization (REQ-026) — I, E	—	Unauthenticated → 401 and the UI redirects to <code>/login</code> ; signed in without either role → 403 and no order data in the response body	TC-114
URL view state (REQ-027) — E	A filtered, sorted, paged view	Reload reproduces it; Back returns to the previous view; a URL with <code>sort=bogus&page=-1</code> renders the default view without an error page	TC-115
Empty and no-match states (REQ-020, REQ-022) — E	Empty table / non-matching filter	MSG-I003 with New production order / MSG-I004 with Clear filters , and the filter panel keeps its values	TC-116
Query failure (REQ-020) — E	API returns 500	MSG-E013 banner with Retry ; Retry re-runs the same query	TC-117
Accessibility — U (axe), E (axe)	Table rendered	No axe violations; <code>aria-sort</code> on the active column only; table has a caption; every row reachable by keyboard; summary announces the total	TC-118
Index usage (002_DB) — I	Seeded data	The default-sort and fragment queries use <code>ix_production_orders_due_date_order_number</code> and the trigram index (<code>EXPLAIN</code>), not a sequential scan	TC-119

Unresolved decisions: none for the design. Two items are recorded as limitations rather than decisions:

- The list endpoint reuses the `ProductionOrderEditor` policy, so a read-only role cannot be expressed. This waits on WI-001 DEC-015 (the full role/permission matrix), which remains open.
- The mockup is a static rendering, not a prototype: it fixes layout, state coverage and the treatment of status, the overdue marker and the sort indicator, but no interaction is demonstrated there — behavior is specified in 002_DD-SPD.